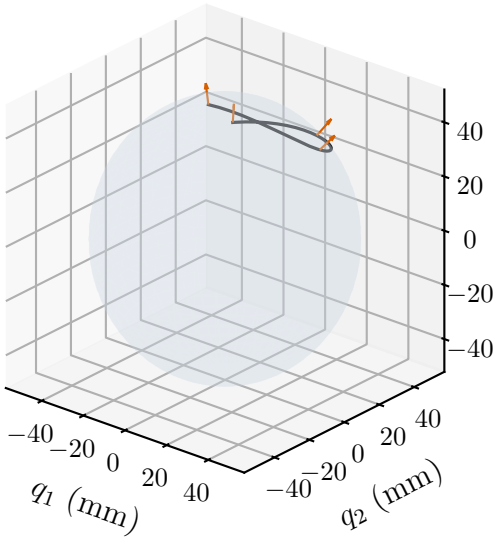
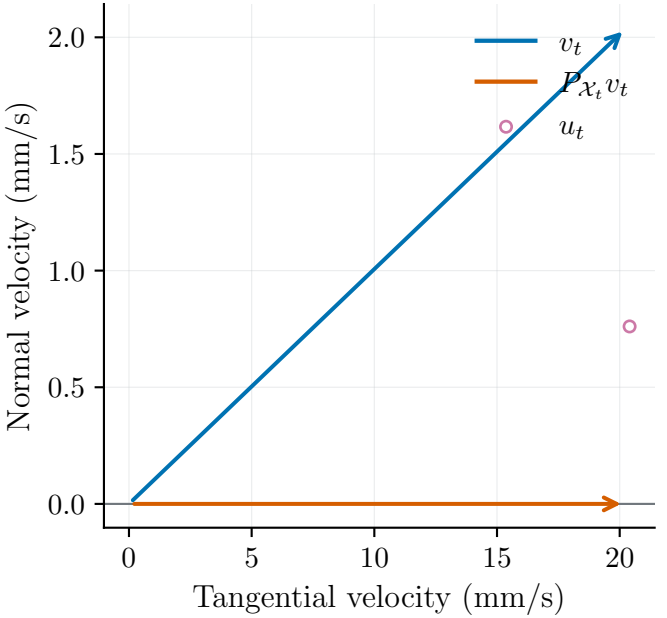

40
20
0
−20
−40
−40
−20
0
20
40
q_1 (mm)
−40
−20
0
20
40
q_2 (mm)
q_3 (mm)


2.0
1.5
1.0
0.5
0.0
Normal velocity (mm/s)
0
5
10
15
20
Tangential velocity (mm/s)
v_t
$P_{\mathcal{X}_t} v_t$
u_t